Rayyan Khan

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Education

Texas A&M University Expected Graduation: December 2027

Bachelor of Science in Computer Engineering

- Relevant Coursework: Computer Architecture, Data Structures and Algorithms, Object Oriented Programming, Artificial Intelligence, Distributed Systems, Operating Systems, Concurrency

Experience

Walmart Global Tech June 2026 – August 2026

Software Engineer Intern

- Engineered a data observability engine in Apache Spark to validate 660M+ daily records, ensuring 0 downstream issues
- Built a Go MCP server enabling AI agents to diagnose pipeline failures via natural language, reducing query time by 70%
- Implemented semantic search over a validation metrics vector store, cutting on-call investigation from hours to minutes
- Designed extensible validation framework for BigQuery datasets, enabling 3+ teams to onboard across 20+ pipelines

HubSpot January 2026 – May 2026

Software Engineer Co-op

- Reduced word error rate 18% across Google and OpenAI speech models, improving accuracy for 1M+ monthly recordings
- Built audio segmentation pipeline for long recordings, improving transcription accuracy 26% for HubSpot's AI Notetaker
- Engineered Kafka pipeline handling 25k+ events/sec, delivering real-time transcripts across thousands of daily meetings
- Scaled transcription infrastructure on Java, HBase, and AWS processing 10 million+ customer call recordings monthly

Schweitzer Engineering Laboratories June 2025 – September 2025

Software Engineer Intern

- Optimized C++ protective relay firmware using Boost and multithreading, achieving 15% faster fault detection times
- Built Jenkins CI/CD framework catching 90% of bugs pre-production, accelerating firmware releases for utility customers
- Engineered Redis caching layer, improving field engineer diagnostic speed by 20% via a 40% query latency reduction
- Designed Kafka/Spark streaming pipeline with PostgreSQL, processing 1TB+ monthly data 36% faster

Texas A&M University June 2024 – September 2024

Undergraduate Researcher

- Boosted medical imaging workflow efficiency with Python and OpenCV, achieving 30% higher tumor detection accuracy
- Collaborated with PhD researchers to debug outputs using NumPy and Matplotlib, improving iteration speed by 25%
- Optimized PyTorch model inference with CUDA GPU acceleration, reducing clinical deployment latency by 44%
- Architected scalable data pipeline using AWS Lambda and S3, processing 100GB+ medical images 50% faster

Projects

Electric Scooter Audio Engine System | *C++, Raspberry Pi, Linux*

- Transformed an electric scooter into a Lamborghini SVJ, synthesizing real-time engine sounds via a C++ audio engine
- Designed 6-state audio state machine with seamless crossfading across the full throttle range
- Built SPI throttle signal pipeline with 3ms end-to-end latency, translating hardware input to audio output in real time
- Designed a custom PCB and configured Linux ALSA stack, managing low-level driver interfaces for real-time audio output

Solana Cryptocurrency Copy Trading Platform | *C++, Scala, Django*

- Engineered low-latency C++ monitoring service achieving 100ms-400ms transaction copy times via gRPC
- Implemented LRU cache for liquidity pool data, reducing RPC calls by 75% and improving speed by 50%
- Designed parallel transaction builder for multiple RPC functions, leading to 43% faster execution times
- Scaled platform to 300+ beta users generating $10,000+ monthly volume with Django/Scala backend

Skills

Programming Languages: Python, Java, C/C++, Golang, Scala, SQL
Frameworks & Libraries: React, Django, NodeJS, Boost, OpenCV, PyTorch, GraphQL, Redis, Socket.io, NumPy
Tools: AWS, Docker, Kubernetes, Jenkins, Git, Kafka, Spark, MongoDB, PostgreSQL, DynamoDB
Concepts: Distributed Systems, Multithreading, CI/CD, REST/GraphQL, Microservices, Generative AI